

JASON BOWMAN

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Electrical Engineer specializing in analog/digital circuit design and control system architecture. Proven ability to own hardware projects from initial concept through prototype testing, validation, and deployment while communicating technical ideas clearly across teams.

EDUCATION

Harvey Mudd College – Claremont, CA

May 2025

Bachelor of Science in Engineering, Concentration in Russian

SKILLS: Power Electronics, Altium, KiCad, Circuit/EM Simulation (SPICE, Sonnet), Hardware Debug & RCA, VNA & Spectrum Analyzer, Control Systems, FPGA (SystemVerilog) & Firmware, MATLAB, C/C++, Python, SolidWorks

EXPERIENCE

Rainmaker – El Segundo, CA

Jul 2025 – Present

Electrical Engineer

- Designed, laid out, and validated flare ignition control, sensing, and protection hardware for cloud seeding
- Isolated UAS thermal and power failures with a 20 kW DC load, driving key electrical and mechanical redesigns
- Built a battery-powered, prop-mounted wireless datalogger, sharpening de-icing system models and iteration
- Evaluated 24 GHz FMCW leakage cancellation and proposed a temperature-stable PA to limit LNA compression
- Modeled a Klopfenstein antenna taper and Sonnet-simulated CPWG to minimize reflections and crosstalk

Sokil Clinic Project (Drone-Based Non-Ferrous Landmine Detection) – Claremont, CA

Aug 2024 – May 2025

Team Lead & Engineer

- Led a 6-person research team to identify the 3 most viable sensor solutions for target landmine characteristics
- Developed RADAR, LiDAR, and multispectral testing, attaining 70%+ detection rates for RADAR and LiDAR
- Co-authored a 60-page technical report that shaped the sponsor's strategic direction

Microprocessor Systems Design & Application – Claremont, CA

Aug 2024 – Dec 2024

Embedded Systems Engineer

- Designed MCU (STM32) and FPGA hardware/software for a squat-tracking biomechanical device with feedback
- Used waveform software and logic analyzers to debug and verify I2C communication
- Co-developed 2000+ lines of C and SystemVerilog code with a partner using GitHub
- Provided real-time squat form feedback to the user through a custom VGA driver

High Power Rocketry – Claremont, CA

Jan 2024 – May 2024

Control Systems Engineer

- Implemented real-time data collection and filtering of 12 flight parameters, enabling spatial awareness during flight
- Tuned a PID feedback loop using the Ziegler-Nichols method, resulting in stable rocket flight
- Designed and simulated a 3D-printed electronics mount in SolidWorks to fit body tolerances and survive 30 G shocks

LEADERSHIP EXPERIENCE

Second Lieutenant - US Army Reserves (Clearance Level: Secret) – Fallbrook, CA

Staff Section Officer In Charge

Mar 2026 – Present

- Manage operations for 7 enlisted personnel, providing strategic insight to unit commanders
- Plan and coordinate operations, cross-functional training, and briefings to inform executive action

Claremont-Mudd-Scripps Track & Field – Claremont, CA

Team Captain

Aug 2024 – May 2025

- Balanced demanding STEM coursework while achieving All-Region honors
- Mentored underclassmen CMS athletes to set goals and achieve academic and athletic success

LANGUAGES

- **Languages:** English (Native), Russian (Intermediate/OPIc 1+)